

Genesis Protect Acute v1

Claim Verification Demo Trace

8-Step Proof: claim data hashed, verified, connected to coverage decision

Program ID: Bn6eixac1QEEVERGBvBjxAd6pgB9e2q4XHvAkinQ5y1B

SCENARIOS COVERED

EVT7-T1-001	Event 7	Tier 1	Heat stroke, Las Vegas NV	APPROVED	\$300
TRV30-T3-001	Travel 30	Tier 3	Ischemic stroke, New York NY	APPROVED	\$3,000
TRV30-T3-DEN-001	Travel 30	Tier 3	Elective bariatric, New Delhi	DENIED	(fraud)

OmegaX Protocol | KR 1.5 Technical Walkthrough | May 2026

Fully Simulated — No Network Calls | Internal — Pre-Mainnet

How the 8-Step Verification Chain Works

#	Action	On-Chain Effect
1	Policy chain verification	Confirm PolicySeries 5-hash fields + OutcomeSchema.verified
2	Evidence packet hashing	Build canonical JSON -> SHA-256 -> evidence_ref_hash
3	Claim case init	ClaimCase PDA derived; state = OPEN (0)
4	Oracle submission	State: OPEN -> UNDER_REVIEW (1)
5	Hash anchor	evidence_ref_hash written to ClaimCase PDA; immutable
6	Tier classification	Operator classifies tier; decision_support_hash computed
7	Oracle attestation	ClaimAttestation PDA; attestation_ref_hash == evidence_ref_hash enforced
8	Coverage decision	State -> APPROVED (2) / DENIED (3); obligation path or void

Security invariant: `attestation_ref_hash` must equal `evidence_ref_hash` — enforced on-chain at **attest_claim_case**. Transaction reverts on mismatch. PHI never touches Solana; only 32-byte SHA-256 hashes are anchored.

SCENARIO EVT7-T1-001

Product: EVENT 7 (7-day fixed benefit) | Tier 1: ER_SAME_DAY | Outcome: APPROVED

Patient	34y M US	Location	Las Vegas, NV
Pre-conditions	none	Facility	University Medical Center of Southern Nevada
Diagnosis	Exertional heat stroke	Coverage day	1

1 Policy Chain Verification

Field	Value
PolicySeries address	6ZfyGQUcW132mEmYBmT5RtoagZyTHi2gTuGQUHW2qTLX
material_locked	True
terms_hash	a7f2c4e1d9b3a8f...8f5c3e7d1b4a9f6
pricing_hash	b3c8e5f1a9d2c7e...1b8a4d7c9e2f5a1
payout_hash	c5e1b7d3a9f4c2e...2b8d5a1c7e4b3d9
reserve_model_hash	d2b9f5c1e7a3d8b...4b7f1c5e9a2d3b8
evidence_req_hash	e8d4c2b7f1a5e3d...7c9e2b4a6d8c1f5
HealthPlan.schema_binding_hash	e9c1f4a7d3b6e2c...5b3f8a1d6c4e7b9
OutcomeSchema.verified	True

2 Evidence Packet Construction & SHA-256

Field	Value
schema	genesis-protect-acute-evidence-v1
claim_id	evt7-t1-sim-001
member wallet	sim-member-wallet-EVT7-T1-001
ICD-10	T67.01XA
docs hashed	3 files (SHA-256 per document)
packet size	1000 bytes (canonical JSON)

evidence_ref_hash — SHA-256 of canonical evidence packet

b19efc90ff8c147d1ac61bf94b550ffb09d26f3b1e7f7b312d824516
da8d6ca4

3 Claim Case PDA Initialization

Field	Value
PDA seeds	"claim_case" health_plan claim_id
ClaimCase PDA	3giTWacLpUrruRFtMyB7jk3582aeM6TPVrSQKiFFprBd
initial state	0 — OPEN
policy_series	6ZfyGQUcW132mEmYBmT5RtoagZyTHi2gTuGQUHW2qTLX

4 Oracle Submission (OPEN -> UNDER_REVIEW)

Field	Value
state transition	OPEN (0) -> UNDER_REVIEW (1)
oracle operator	BGN6pVpuD9GPSsExtBi7pe4RLCJrkFVsQd9mw7ZdH8Ez
documents	3 items submitted
review SLA	24 h target

5 Evidence Hash Anchor (On-Chain)

evidence_ref_hash anchored to ClaimCase PDA. Immutable once attestation_count ≥ 1. Raw PHI never written to Solana.

evidence_ref_hash — anchored at ClaimCase PDA (immutable)

b19efc90ff8c147d1ac61bf94b550ffb09d26f3b1e7f7b312d824516
da8d6ca4

6 Clinical Tier Classification & Decision Support

Field	Value
tier classified	TIER_1_ER_SAME_DAY
review outcome	APPROVED
review hours	3.5 h

decision_support_hash — SHA-256 of operator review bundle

7dc989f2deef053a6fbc6436dda509544e37382815b4f3a859d992a
067e2cc6

7 Oracle Attestation (ClaimAttestation PDA)

Field	Value
PDA seeds	"claim_attestation" claim_case oracle
ClaimAttestation PDA	Atcj4U8v8PygSQki7YzG8mWSP9Yo25xwLgWhwCsdscav
attestation_count	1
attestation_ref_hash == evidence_ref_hash	MATCH (tx proceeds)

attestation_ref_hash == evidence_ref_hash (on-chain enforced)

b19efc90ff8c147d1ac61bf94b550ffb09d26f3b1e7f7b312d824516
da8d6ca4

8 Coverage Decision & Settlement

APPROVED

Settlement: \$300 USDC

Funding line: genesis-event7-premiums

Field	Value
final claim state	4 — SETTLED
approved amount	\$300.00 USDC
protocol fee	50 BPS = \$1.00 USDC
member payout	\$299.00 USDC

funding line	genesis-event7-premiums
funding line addr	2115rGD6zKmUhLGk9zwqbA9tdcA5nuAwRNAQDLcgSpWA
obligation path	PROPOSED -> RESERVED -> CLAIMABLE_PAYABLE -> SETTLED

SCENARIO TRV30-T3-001

Product: TRAVEL 30 (30-day hybrid) | Tier 3: SURGERY_ICU_2NIGHTS | Outcome: APPROVED

Patient	67y F FR	Location	New York, NY
Pre-conditions	atrial fibrillation (apixaban), hypertension, hyperlipidemia	Primary	NewYork-Presbyterian / Weill Cornell
Diagnosis	Acute ischemic stroke (left MCA, cardioembolic) — CVA	Warranty	3

1 Policy Chain Verification

Field	Value
PolicySeries address	29XmfdHceAeAvtiESAcNDXLsJxEqW2RBa3DttTUUcco
material_locked	True
terms_hash	f1a6d3c8e2b5f9a...5a9d2c7e1b6f3a8
pricing_hash	a2e7b4d9c5f1a8e...3e8b2d5c1f9a4e7
payout_hash	b8f3a5d1c9e6b2f...3f9a2d7c5e4b6f8
reserve_model_hash	c4e9b1f6a3d8c5e...8e5b9f7a4d3c1e6
evidence_req_hash	d7a2f5c8e1b4d6a...9a5f3c1e8b2d4a7
HealthPlan.schema_binding_hash	e9c1f4a7d3b6e2c...5b3f8a1d6c4e7b9
OutcomeSchema.verified	True

2 Evidence Packet Construction & SHA-256

Field	Value
schema	genesis-protect-acute-evidence-v1
claim_id	trv30-t3-sim-001
member wallet	sim-member-wallet-TRV30-T3-001
ICD-10	I63.30
docs hashed	5 files (SHA-256 per document)
packet size	1297 bytes (canonical JSON)

evidence_ref_hash — SHA-256 of canonical evidence packet

2b7ad70522338fd363215e80f858ac1639fbf045a15b5610739dfedf
c9b984b0

3 Claim Case PDA Initialization

Field	Value
PDA seeds	"claim_case" health_plan claim_id
ClaimCase PDA	6wwFAD4xmxWutRaFW5zn25oaTZbBdefXJ8gqn1HtkfK6
initial state	0 — OPEN
policy_series	29XmfdHceAeAvtiESAcNDXLsJxEqW2RBa3DttTUUcco

4 Oracle Submission (OPEN -> UNDER_REVIEW)

Field	Value
state transition	OPEN (0) -> UNDER_REVIEW (1)
oracle operator	BGN6pVpuD9GPSsExtBi7pe4RLCJrkFVsQd9mw7ZdH8Ez
documents	5 items submitted
review SLA	24 h target

5 Evidence Hash Anchor (On-Chain)

evidence_ref_hash anchored to ClaimCase PDA. Immutable once attestation_count ≥ 1. Raw PHI never written to Solana.

evidence_ref_hash — anchored at ClaimCase PDA (immutable)

2b7ad70522338fd363215e80f858ac1639fbf045a15b5610739dfedf
c9b984b0

6 Clinical Tier Classification & Decision Support

Field	Value
tier classified	TIER_3_SURGERY_ICU_2NIGHTS
review outcome	APPROVED
review hours	22 h

decision_support_hash — SHA-256 of operator review bundle

95dca162b4662829ba7d801f5c61b443e82b90188ac143bd09110d2b
1b05a556

7 Oracle Attestation (ClaimAttestation PDA)

Field	Value
PDA seeds	"claim_attestation" claim_case oracle
ClaimAttestation PDA	Gy9w924dkdeDdkESE8VV2P3M24mKgPtX7Xyks9GA2uvC
attestation_count	1
attestation_ref_hash == evidence_ref_hash	MATCH (tx proceeds)

attestation_ref_hash == evidence_ref_hash (on-chain enforced)

2b7ad70522338fd363215e80f858ac1639fbf045a15b5610739dfedf
c9b984b0

8 Coverage Decision & Settlement

APPROVED

Settlement: \$3,000 USDC

Funding line: genesis-travel30-liquidity

Field	Value
final claim state	4 — SETTLED
approved amount	\$3,000.00 USDC
protocol fee	50 BPS = \$15.00 USDC
member payout	\$2,985.00 USDC

funding line	genesis-travel30-liquidity
funding line addr	HBrdsf7UjYK5tRoM9j6YaxfV7nFBkVhnJbrmTLVaDiEr
obligation path	PROPOSED -> RESERVED -> CLAIMABLE_PAYABLE -> SETTLED

SCENARIO TRV30-T3-DEN-001

Product: TRAVEL 30 (30-day hybrid) | Tier 3: SURGERY_ICU_2NIGHTS | Outcome: DENIED

Patient	57y M US	Location	New Delhi
Pre-conditions	morbid obesity (BMI 44), type 2 diabetes, sleep apnea, fatty liver	Health plan	Max Super Speciality Hospital Saket
Diagnosis	Laparoscopic sleeve gastrectomy (LSG) — elective bariatric surgery	Episode of care	2

1 Policy Chain Verification

Field	Value
PolicySeries address	29XmfdHceAeAvtiESAcNDXLsJxEqW2RBa3DttTUUcco
material_locked	True
terms_hash	f1a6d3c8e2b5f9a...5a9d2c7e1b6f3a8
pricing_hash	a2e7b4d9c5f1a8e...3e8b2d5c1f9a4e7
payout_hash	b8f3a5d1c9e6b2f...3f9a2d7c5e4b6f8
reserve_model_hash	c4e9b1f6a3d8c5e...8e5b9f7a4d3c1e6
evidence_req_hash	d7a2f5c8e1b4d6a...9a5f3c1e8b2d4a7
HealthPlan.schema_binding_hash	e9c1f4a7d3b6e2c...5b3f8a1d6c4e7b9
OutcomeSchema.verified	True

2 Evidence Packet Construction & SHA-256

Field	Value
schema	genesis-protect-acute-evidence-v1
claim_id	trv30-t3-den-001
member wallet	sim-member-wallet-TRV30-T3-DEN-001
ICD-10	E66.01 + Z68.44
docs hashed	5 files (SHA-256 per document)
packet size	1357 bytes (canonical JSON)

evidence_ref_hash — SHA-256 of canonical evidence packet

7992a47adbebfdc3fc2a7bc8109df4c6723ccbbaaa49e08514f75efc
3e26200e

3 Claim Case PDA Initialization

Field	Value
PDA seeds	"claim_case" health_plan claim_id
ClaimCase PDA	7copPaMX8HTNMM7qrNQihE28bMKKLujpsyeJGPwH1Ydb
initial state	0 — OPEN
policy_series	29XmfdHceAeAvtiESAcNDXLsJxEqW2RBa3DttTUUcco

4 Oracle Submission (OPEN -> UNDER_REVIEW)

Field	Value
state transition	OPEN (0) -> UNDER_REVIEW (1)
oracle operator	BGN6pVpuD9GPSsExtBi7pe4RLCJrkFVsQd9mw7ZdH8Ez
documents	5 items submitted
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5 Evidence Hash Anchor (On-Chain)

evidence_ref_hash anchored to ClaimCase PDA. Immutable once attestation_count ≥ 1. Raw PHI never written to Solana.

evidence_ref_hash — anchored at ClaimCase PDA (immutable)
7992a47adbebfdc3fc2a7bc8109df4c6723ccbbaaa49e08514f75efc
3e26200e

6 Clinical Tier Classification & Decision Support

Field	Value
tier classified	TIER_3_SURGERY_ICU_2NIGHTS
review outcome	DENIED
review hours	72 h
exclusion clause	genesis-acute-v1 Sec 5.1 (Elective Pre-Planned) + Sec 5.5 (Obesity Treatment) + Sec 8.2 (Fraud/Misrepresentation)

decision_support_hash — SHA-256 of operator review bundle
5d847ed58032f8bdfb15622ae310916e820ce547426bdc412c1f4edd
9ea155d9

7 Oracle Attestation (ClaimAttestation PDA)

Field	Value
PDA seeds	"claim_attestation" claim_case oracle
ClaimAttestation PDA	Hb4tfkLjS7DSfn7LUeP8vRgC6Ji1kG9XGbopn2vtsBBF
attestation_count	1
attestation_ref_hash == evidence_ref_hash	MATCH (tx proceeds)

attestation_ref_hash == evidence_ref_hash (on-chain enforced)
7992a47adbebfdc3fc2a7bc8109df4c6723ccbbaaa49e08514f75efc
3e26200e

8 Coverage Decision & Settlement

DENIED

Zero settlement — claim rejected
Policy voided — fraud referral initiated

Field	Value
final claim state	3 — DENIED
approved amount	\$0
funding line	None — no settlement

obligation	Not created — VOID
denial grounds	genesis-acute-v1 Sec 5.1 (Elective Pre-Planned) + Sec 5.5 (Obesity Treatment) + Sec 8.2 (Fraud/Misrepresentation)
fraud referral	INITIATED — policy voided

Scenario Comparison Summary

Scenario	Product	Tier	Outcome	Settlement	Funding Line	Obligation
EVT7-T1-001	Event 7	T1	APPROVED	\$300.00 USDC	genesis-event7-premiums	SETTLED
TRV30-T3-001	Travel 30	T3	APPROVED	\$3,000.00 USDC	genesis-travel30-liquidity	SETTLED
TRV30-T3-DEN-001	Travel 30	T3	DENIED	\$0.00 USDC	—	VOID

Key takeaways: (1) All three claims traversed the same 8-step chain — the code path is identical; only the oracle outcome differs. (2) Hash integrity is enforced on-chain: attestation_ref_hash must equal evidence_ref_hash or the transaction reverts. (3) PHI separation is maintained throughout: raw medical records never touch Solana. (4) The denial scenario shows that elective pre-planned procedures and fraud are caught at oracle review before any settlement obligation is created. (5) Tier-routed settlement: Event 7 T1 draws from the premiums funding line; Travel 30 T3 draws from the liquidity funding line.